Lab 2.1 TDR from VNA Measurements

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System overview

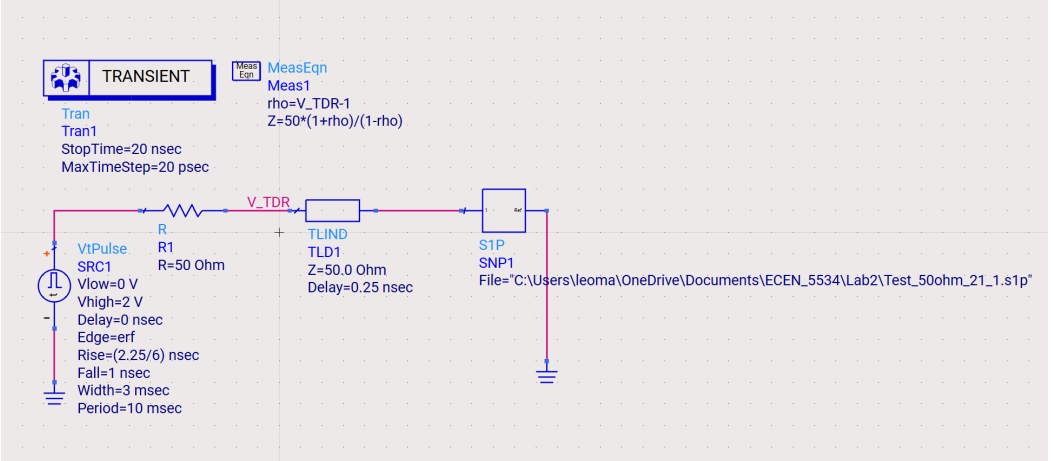
Question: How accurately does a VNA-based virtual TDR measure impedance, and over what range of resistance can we trust it?

DUTs: SMA-mounted resistors from 0 Ω to 1 MΩ and a 50 Ω load

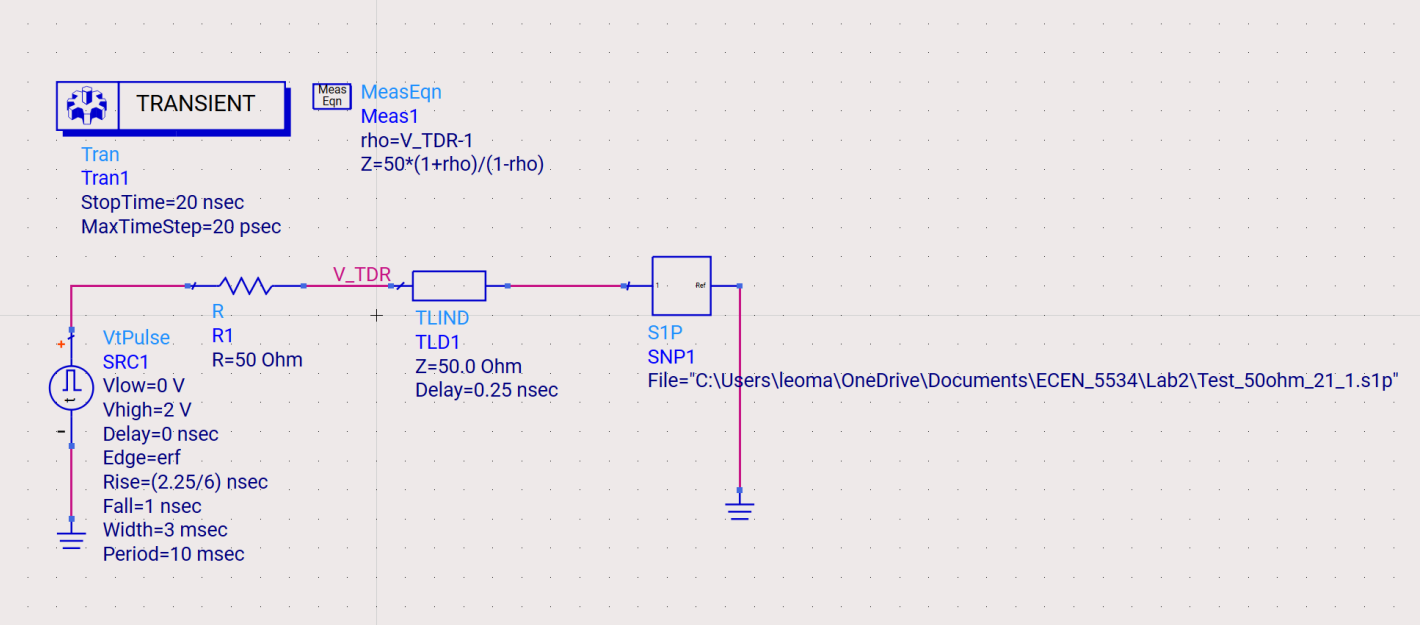
Reference: DMM resistance of every resistor

Instrument: NanoVNA-F V3, 1-port, calibrated at the end of the cable, 801 points, 7.49 MHz start = step, up to 6 GHz

Analysis: .s1p files into an ADS virtual TDR, every value read at 15 ns



How our virtual TDR works



Reflection: the 2 V source drives a matched 50 Ω line, so 1 V is launched and ρ = V_TDR − 1

Impedance: $Z = 50\,(1 + \rho)/(1 - \rho)$

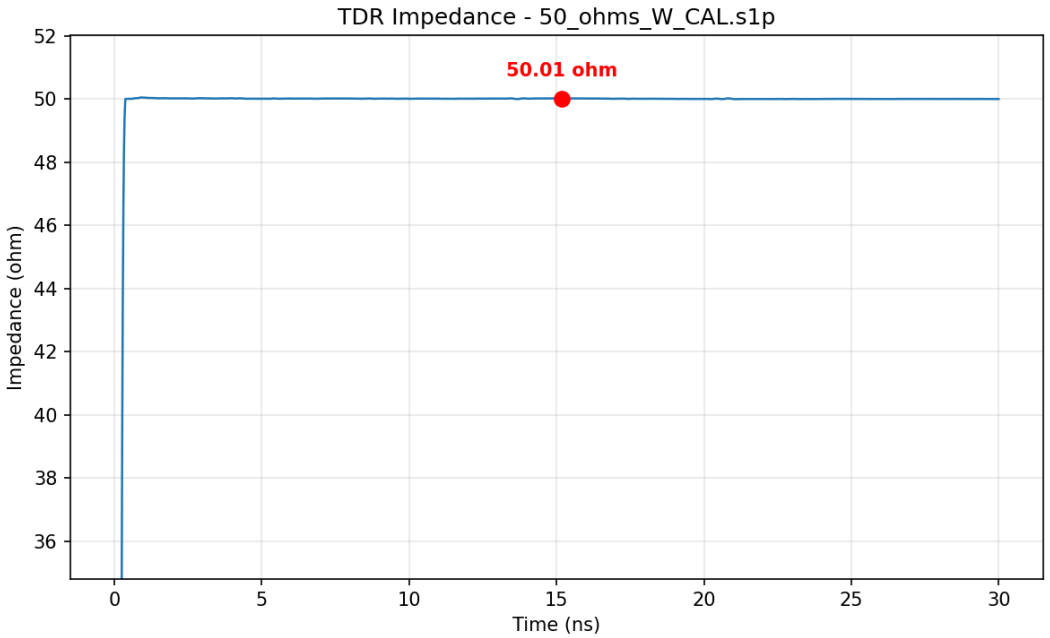
Rise time: erf edge, Rise = 2.25/6 ns, a 10 to 90 rise time of 1/BW ≈ 0.17 ns.

TLIND: 50 Ω, 0.25 ns, gives a flat 50 Ω start before the DUT

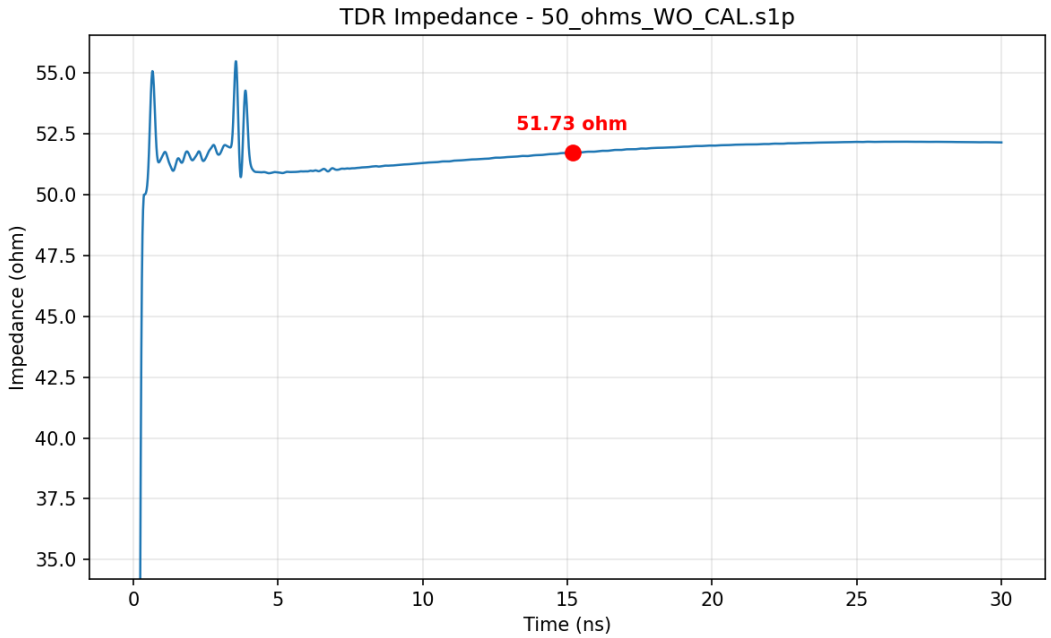
Sweep: start = step = 7.49 MHz, so ADS never interpolates

So what? The measured S11 becomes a behavioral model, so we can build any TDR we want after the measurement, limited only by the VNA bandwidth and frequency step.

Calibrate at the end of the cable



50 Ω load, calibrated at the cable end



Same 50 Ω load, no calibration

Without calibration the reference plane is the VNA port, so the cable is part of the DUT. The load reads 51.7 Ω and drifts from 51 to 52 Ω.

With calibration the same load reads 50.01 Ω and stays flat.

So what? Every impedance we report is referenced to the calibration plane. Without SOLT at the cable end, the cable is in every number.

Resistor sweep: TDR vs DMM

Nominal	DMM (Ω)	TDR, ADS at 15 ns (Ω)	Error (%)
0 Ω	0.10	0.018	below DMM floor
10 mΩ	0.10	0.018	below DMM floor
100 mΩ	0.19	0.101	below DMM floor
1 Ω	1.10	0.970	-11.8*
10 Ω	10.07	9.692	-3.8
100 Ω	100.0	98.10	-1.9
1 kΩ	999	982.7	-1.6
10 kΩ	9,970	**9,403**	**-5.7**
100 kΩ	100,000	**58,604**	**-41**
1 MΩ	1,007,000	**332,778**	**-67**

ADS virtual TDR, marker at 15 ns on every resistor. DMM read the same day. ***DMM includes about 0.1 Ω of lead resistance.***

Accuracy bands:

Within 1%: none. Best is 1 kΩ at -1.6%

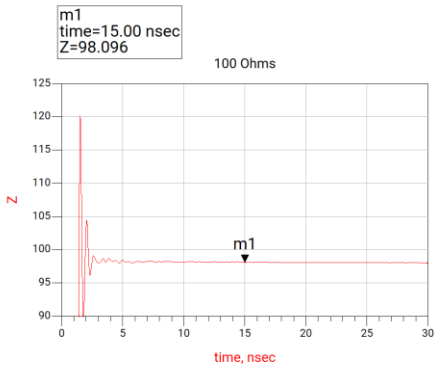
Within 5%: 10 Ω to 1 kΩ

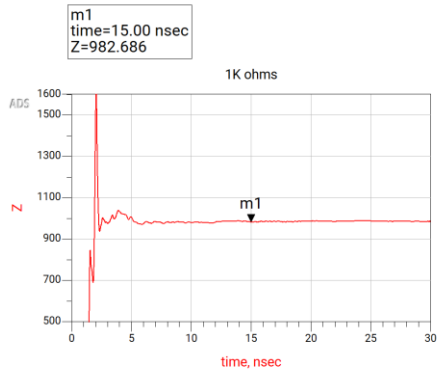
Within 10%: 10 Ω to 1 kΩ. 10 kΩ misses 5% at -5.7%

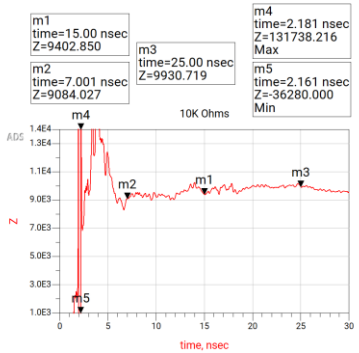
DMM floor: the 0 Ω part reads 0.10 Ω, so DMM values below about 0.2 Ω are mostly lead resistance

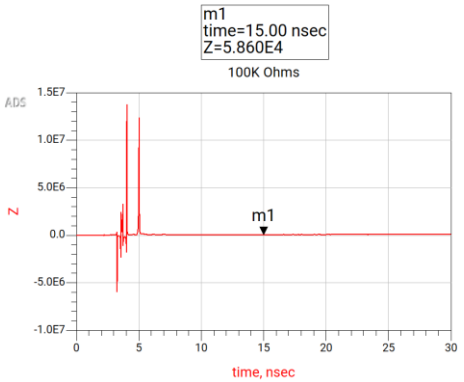
So what? The 1-port TDR is within 5% from 10 Ω to 1 kΩ. It misses 5% at 10 kΩ and is off by 41% at 100 kΩ and 67% at 1 MΩ.

Flat plateaus up to 1 kΩ, wandering above









100 Ω and 1 kΩ settle within ~5 ns and stay flat: 98.1 Ω and 982.7 Ω at 15 ns.

10 kΩ never settles: 9.08 kΩ at 7 ns, 9.40 kΩ at 15 ns, 9.93 kΩ at 25 ns. That is ±0.42 kΩ (±4%) of wander on one sweep.

100 kΩ: the launch spikes reach 10 MΩ and the 15 ns marker reads 58.6 kΩ. 1 MΩ reads 333 kΩ.

Only the resistance changed.

So what? Where the plateau is flat, one marker is enough. At 10 kΩ the wander alone is ±4%, so a single reading cannot be trusted to 5%.

Why 10 kΩ is so sensitive

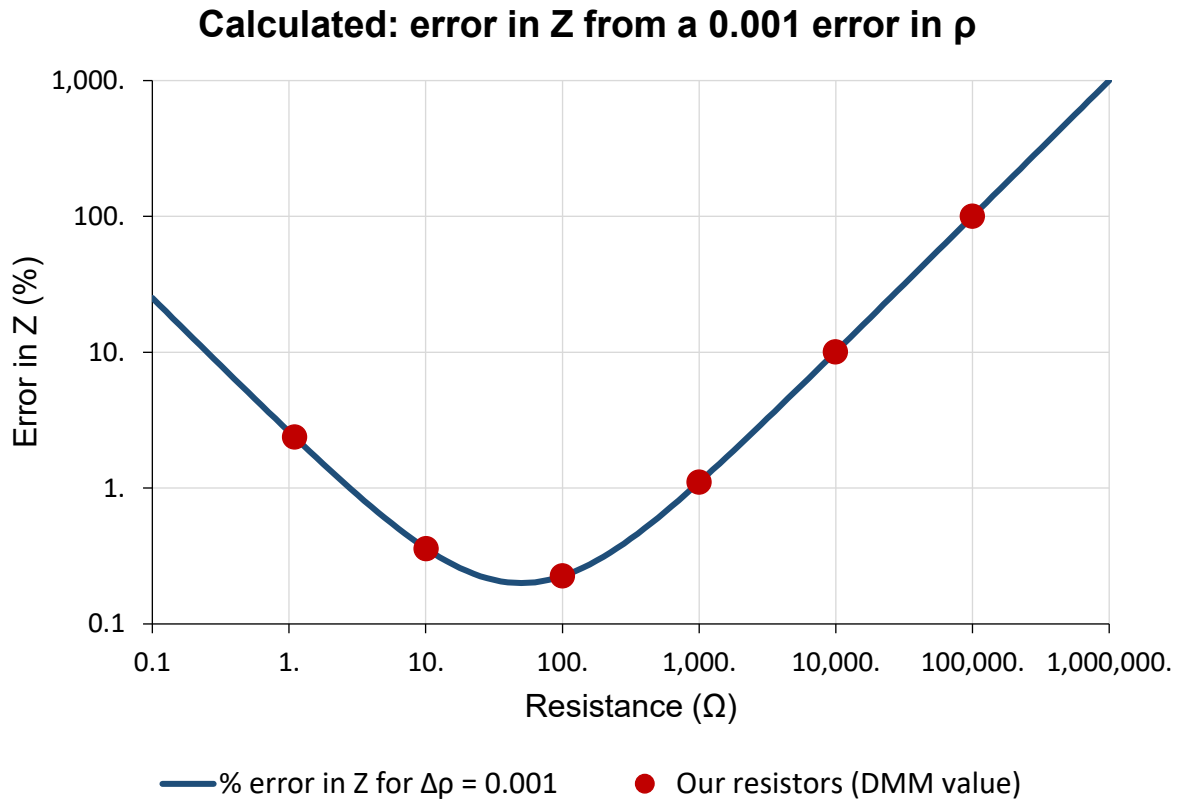
$Z = 50\,(1 + \rho)/(1 - \rho)$ gets steep as $\rho \to 1$

$dZ/d\rho = 100/(1 - \rho)^2\ \Omega$

At 10 kΩ: $\rho = 0.990$, so $dZ/d\rho \approx 1.0\ M\Omega$. Our reading is 570 Ω low, a ρ error of only **0.0006**.

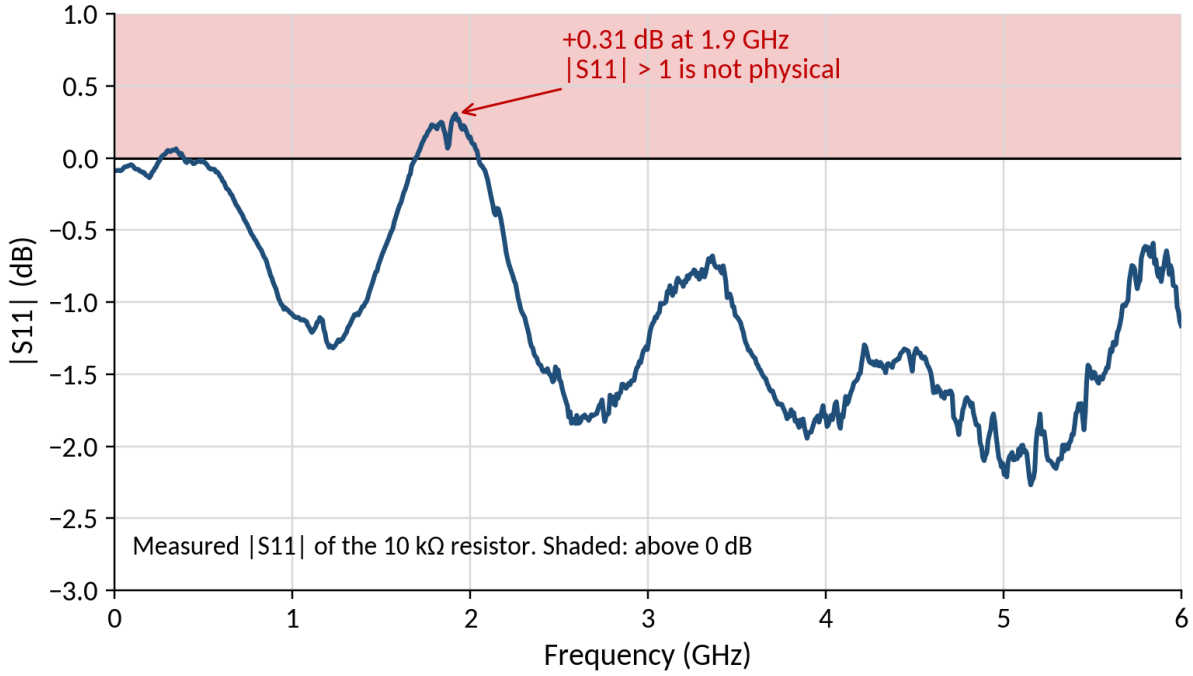
At 50 Ω: $\rho = 0$, the same 0.0006 moves Z by only 0.06 Ω.

Relative error: $\Delta Z/Z = \Delta\rho\,(R + 50)^2/(100\,R)$. Smallest at 50 Ω and the same for R and 2500/R, so 1 Ω and 2.5 kΩ are equally hard.



So what? The 10 kΩ error is not a bad resistor or a bad ADS setup. A tiny error in ρ is amplified about 100× at 10 kΩ versus 2× at 50 Ω.

Our VNA cannot resolve ρ that close to 1



|S11| rises above 0 dB. A passive resistor cannot reflect more than it receives, so the measured ρ is off by at least 0.035 at 1.9 GHz.

Calibration check: At 7.49 MHz the calibrated cable open reads |S11| = 0.9992, a ρ error near 0.001 on a standard we know exactly.

Dynamic range: the NanoVNA-F V3 spec is 50 dB for S11 at 1.5 GHz and 40 dB at 3 GHz. 50 dB is a ρ floor of about 0.003, or about ±3 kΩ at 10 kΩ.

So what? Our 570 Ω miss at 10 kΩ is the size of error the VNA's own ρ resolution predicts, so it is an instrument limit we can quantify, not "measurement error".

Literature: 1-port has a limited Z range

Method	Usable range (Sandler, Table 2)	Covers the 10 kΩ to 1 MΩ values?
1-port reflection (what we used)	1 Ω to 2 kΩ	**No**
2-port series	10 Ω to 2.5 MΩ	Yes
2-port shunt	1 mΩ to 225 Ω	No
Our 1-port TDR, within 5% of DMM	**10 Ω to 1 kΩ**	**No**

Sandler, "Extending the usable range of the 2-port shunt through impedance measurement", IEEE LAMC 2016

Every method loses sensitivity as the S-parameter magnitude nears 1. Limiting it to 0.9, 1-port reflection is good from about 1 Ω to 2 kΩ.

Our measured 5% range, 10 Ω to 1 kΩ, sits inside the paper's 1-port range. Our 10 kΩ, 100 kΩ and 1 MΩ are past it.

So what? To measure 10 kΩ well, change the method (2-port series), not the post-processing. We only did 1-port measurements in this lab.

So what?

A VNA plus ADS is a working TDR. **The 50 Ω cal load reads 50.01 Ω** and resistors from 10 Ω to 1 kΩ read within 5% of the DMM.

Our 1-port VNA is limited. **10 kΩ reads 5.7% low, 100 kΩ 41% low and 1 MΩ 67% low**, because $Z = 50(1 + \rho)/(1 - \rho)$ amplifies tiny ρ errors as $\rho \rightarrow 1$.

The takeaway is to know the impedance range your instrument is accurate to before you trust a number from it. Outside about 10 Ω to 1 kΩ, our 1-port VNA needs a 2-port method.

Thank you

Any Questions?